Appendix A
Informed Consent Form / Memorandum of Agreement

MEMORANDUM OF AGREEMENT AND INFORMED CONSENT FOR RESEARCH PARTICIPATION

This Memorandum of Agreement and Informed Consent for Research Participation is entered into on this 11th day of October, 2026, by and between:

University of San Carlos – Computer Engineering, represented by the student researchers:

Paul Andrew F. Parras
Vaun Michael C. Pace
Lorenz Anthony L. Soriano

hereinafter referred to as the **"Researchers,"**

and

Lorenz Go Lim / LGL PIGGY FARM, located at Santa Cruz, Tagoloan, Misamis Oriental,
hereinafter referred to as the **"Client/Participant."**

1. PURPOSE AND SCOPE

This Agreement establishes the collaboration between the Researchers and the Client for the design, development, installation, and testing of a piggery monitoring and automation system as part of an academic thesis project entitled **"Integrated Automated Feeding and Watering System with Centralized Data Monitoring for Commercial Piggeries."**

The system to be developed is a prototype that may include feeding and drinking mechanisms, monitoring components, and data collection features. It will be installed and tested within the Client's piggery environment to observe performance, gather data, and evaluate feasibility.

The primary objective of the project is to explore a cost-effective and efficient approach to reducing manual labor and improving monitoring of pig behavior and farm conditions. The system is intended solely for academic and research purposes and is not designed for immediate commercial deployment.

The scope of this collaboration includes on-site installation, periodic monitoring, data gathering, system adjustments, and evaluation activities during the testing phase. Any system specifications or features may be refined based on the Client's operational needs and practical constraints.

2. INVITATION AND PARTICIPATION IN THE STUDY

The Client is being invited to participate because the piggery provides an appropriate environment for the installation and testing of the prototype system. Participation may involve providing access to the piggery, coordinating testing activities, becoming familiar with the system, and providing feedback regarding its usability and functionality.

As part of the evaluation of the system, the Client or designated caretaker may be asked to participate in user acceptance testing (UAT), complete an evaluation checklist, and provide responses through a five-point Likert-scale evaluation concerning the usability, functionality, monitoring features, alerts, feeding schedule configuration, and perceived usefulness of the system.

Participation in these evaluation activities is voluntary.

3. WHAT WILL HAPPEN DURING PARTICIPATION

If the Client or designated caretaker agrees to participate, the following activities may be conducted:

1. Provide the Researchers with access to designated areas of the piggery required for installation and testing.
2. Coordinate with the Researchers regarding installation, testing, monitoring, and maintenance schedules.
3. Observe or interact with the system dashboard and monitoring interface.
4. Review system information such as feeding-related data, water-level information, air-quality alerts, and system notifications.
5. Participate in user acceptance testing of the system.
6. Complete an evaluation checklist regarding the system's usability and functionality.
7. Complete a five-point Likert-scale evaluation regarding the system, including aspects such as interface clarity, monitoring information, alerts, feeding schedule configuration, and perceived reduction in manual monitoring effort.
8. Provide comments or feedback regarding the system's operation and suitability for the piggery.

The Researchers may also record relevant system events and operational observations during the testing period for academic analysis and documentation.

No medical procedures, invasive procedures, or procedures involving physical intervention on the Client or caretaker will be performed as part of the study.

4. EXPECTED DURATION OF PARTICIPATION

The Agreement shall take effect on **[Start Date]** and remain valid until **[End Date]** or until the completion of the thesis project, subject to the terms stated in this document.

The Client or designated caretaker's participation in evaluation activities is expected to involve approximately **[insert estimated duration]** per evaluation session and **[insert number of sessions]** evaluation session(s), depending on the testing schedule and project requirements.

The Researchers shall coordinate the schedule of activities with the Client to minimize disruption to normal farm operations.

5. EXPERIMENTAL NATURE OF THE SYSTEM

The system is a **prototype developed for academic research** and may have limitations in terms of accuracy, reliability, and performance. It is not guaranteed to fully automate farm operations or prevent livestock-related issues.

The system is intended only to assist in monitoring and basic automation and should not be relied upon as a sole decision-making tool.

The testing and evaluation of the prototype are experimental in nature and are intended to determine its performance, usability, functionality, and feasibility within the Client's piggery environment.

6. POSSIBLE DISCOMFORTS AND INCONVENIENCES

Participation is not expected to involve significant physical discomfort. However, participation may require time from the Client or designated caretaker for system familiarization, user acceptance testing, evaluation forms, feedback, and coordination with the Researchers.

Installation, testing, or maintenance activities may also cause temporary inconvenience or minor disruption to normal farm operations.

The Researchers shall coordinate activities with the Client and conduct them in a manner intended to minimize unnecessary disruption.

7. POSSIBLE RISKS

The study does not involve medical procedures or invasive procedures involving human participants.

Potential risks associated with participation may primarily involve minor inconvenience, interruption of normal farm activities, or possible dissatisfaction with the prototype's performance.

Because the system is an experimental prototype, there may also be limitations in accuracy, reliability, or performance. The Researchers will implement the system with consideration for basic safety practices and will monitor the system during testing.

The prototype should not be relied upon as the sole basis for decisions concerning farm operations or livestock management.

8. RANDOM ASSIGNMENT

Participants will **not be randomly assigned** to different treatment groups or experimental groups. Participation consists of evaluating and providing feedback on the prototype system within the Client's piggery environment.

9. REIMBURSEMENT AND COMPENSATION

There is **no monetary compensation or reimbursement currently planned for participation** in the study.

Participation in the study is voluntary, and the Client or designated caretaker will not be required to incur expenses specifically for participating in the research evaluation activities.

If any changes regarding reimbursement or compensation arise, these shall be communicated to the participant before the activity concerned.

10. POSSIBLE BENEFITS TO THE PARTICIPANT

The Client or designated caretaker may benefit from becoming familiar with a prototype system intended to assist with feeding, watering, monitoring, and centralized data management.

The study may also provide the Client with an opportunity to provide practical feedback that may help improve the usability and functionality of the proposed system.

However, the Researchers do not guarantee that the Client or caretaker will receive a direct or immediate benefit from participation.

11. POSSIBLE BENEFITS TO THE COMMUNITY, STAKEHOLDERS, AND SCIENTIFIC KNOWLEDGE

The study may contribute to the development and evaluation of technology intended to improve monitoring and automation practices in commercial piggery environments.

The results may provide information regarding the feasibility of integrating automated feeding and watering mechanisms with centralized monitoring and data collection.

The findings may also serve as a reference for future academic research and development involving agricultural automation, livestock monitoring, embedded systems, and centralized data monitoring.

12. ALTERNATIVES TO PARTICIPATION

The Client may choose not to participate in the research activities. There are no alternative experimental treatments or procedures that the Client is required to undergo.

Choosing not to participate will not result in a penalty or loss of benefits to which the Client is otherwise entitled.

13. COMPENSATION OR MEDICAL TREATMENT IN CASE OF INJURY

The study does not involve medical procedures or procedures intended to cause physical intervention on human participants.

In the unlikely event that an injury occurs to a participant in connection with an activity directly conducted by the Researchers, the incident shall be documented and appropriately referred for necessary assistance in accordance with the applicable policies of the University of San Carlos and the approved research protocol.

No specific medical treatment or monetary compensation is guaranteed by this Agreement unless otherwise required by applicable institutional policy or approved research arrangements.

14. RESPONSIBILITIES OF THE RESEARCHERS

The Researchers shall be responsible for the design, development, installation, and testing of the system. They will ensure that all components are properly set up and functioning during the project period.

The Researchers shall conduct monitoring, collect relevant data, and perform necessary adjustments or troubleshooting as needed.

They shall also ensure that the system is implemented in a manner that minimizes disruption to normal farm operations and adheres to basic safety practices.

15. RESPONSIBILITIES OF THE CLIENT

The Client agrees to provide the Researchers with access to the piggery and allow installation and testing of the system within the premises.

The Client shall provide the necessary electrical power for system operation and cooperate by sharing relevant operational information when required.

The Client also agrees to coordinate with the Researchers regarding schedules and activities related to the project.

The Client or designated caretaker may participate in the evaluation activities described in this document and may provide feedback regarding the system's usability and functionality.

16. USE OF FACILITIES AND RESOURCES

The Client permits the use of available space and electrical power within the piggery for the duration of the study.

The Researchers agree to use all facilities and resources responsibly and avoid excessive or unnecessary consumption.

Any installation shall be done with consideration for the existing setup of the farm.

17. SYSTEM NATURE AND LIMITATIONS

The system is a **prototype developed for academic research** and may have limitations in terms of accuracy, reliability, and performance.

It is not guaranteed to fully automate farm operations or prevent livestock-related issues.

The system is intended only to assist in monitoring and basic automation and should not be relied upon as a sole decision-making tool.

18. LIABILITY

The Researchers shall take reasonable measures to conduct installation and testing in a safe and responsible manner.

The system is experimental in nature and may have limitations or malfunctions during testing. The Researchers shall monitor and troubleshoot the system as necessary.

The Client acknowledges the prototype nature of the system and the limitations associated with its testing and operation.

Any loss, damage, or injury arising during the study shall be documented and addressed in accordance with applicable University policies and the approved research protocol.

19. DATA, PRIVACY, AND CONFIDENTIALITY

All data collected during the project shall be used strictly for academic purposes, including analysis, documentation, and presentation.

The Researchers agree to maintain the confidentiality of sensitive information related to the Client's operations and shall not disclose such information without appropriate permission.

Information obtained from the Client or piggery that is not necessary for the research objectives shall not be intentionally collected.

Evaluation responses, feedback, and other information provided by the Client or caretaker shall be treated as research data and shall be used only for the purposes described in this document.

The Researchers shall take reasonable measures to prevent unauthorized disclosure of participant and Client information.

20. RECORDING OF EVALUATION ACTIVITIES

The Researchers may record relevant system events, testing results, operational observations, UAT results, evaluation responses, and feedback for purposes of research analysis and thesis documentation.

Any recording or collection of information involving the Client or designated caretaker shall be limited to information relevant to the study.

No audio, video, or photographic recording of the participant will be conducted unless separately explained and permission is obtained.

21. VOLUNTARY PARTICIPATION AND RIGHT TO WITHDRAW

Participation in the research evaluation activities is voluntary.

The Client or designated caretaker may refuse to participate in any evaluation activity or may withdraw from participation at any time without penalty, discrimination, or loss of benefits to which they are otherwise entitled.

Choosing not to participate or withdrawing from the evaluation activities will not affect the Client's existing relationship with the University of San Carlos or the Researchers.

The Client may also request clarification regarding any part of the study before deciding whether to participate.

22. TERMINATION OF THE AGREEMENT

This Agreement shall remain valid until **[End Date]** or until completion of the thesis project, unless terminated earlier.

Either party may terminate the Agreement with prior notice if necessary.

The Researchers may also discontinue testing if continued operation of the prototype presents a safety, operational, technical, or other significant concern.

In the event of termination, appropriate arrangements shall be made regarding the system, installed components, collected data, and any remaining project activities.

Termination of the research participation will not result in a penalty to the Client or designated caretaker.

23. SYSTEM REMOVAL OR TURNOVER

Upon completion of the study, the Researchers shall remove the system from the Client's premises unless both parties agree to leave or transfer the system to the Client.

Any such arrangement shall be mutually agreed upon.

24. CONTACT INFORMATION

For questions regarding the study, participation, evaluation activities, or the operation of the research prototype, the participant may contact:

Paul Andrew F. Parras
Student Researcher
Contact Number: (+63) 906 394 8224
Email Address: parraspaul13874@gmail.com

Vaun Michael C. Pace
Student Researcher
Contact Number: (+63) 961 241 0607
Email Address: pacevaun@gmail.com

Lorenz Anthony L. Soriano
Student Researcher
Contact Number: (+63) 917 726 0812
Email Address: lorenzal.soriano@gmail.com

For questions regarding the research supervision, the participant may contact:

Engr. Alvin Joseph S. Macapagal
Thesis Adviser
Contact Number/Email: alvinjoseph.macapagal@usc.edu.ph

Engr. Jan Dave Q. Campañera
Assistant Thesis Adviser
Contact Number/Email: jandave.campanera@usc.edu.ph

25. RESEARCH ETHICS COMMITTEE REVIEW AND PARTICIPANT CONCERNS

This research project will be submitted to the appropriate **University of San Carlos Research Ethics Committee (REC)** for ethical review in accordance with applicable institutional requirements.

If the participant has concerns regarding their rights as a research participant, the participant may contact the University of San Carlos Research Ethics Committee through:

Research Ethics Committee
University of San Carlos

The participant may use this contact information to raise questions or concerns regarding participation in the research.

CONSENT DECLARATION

I have read and understood the information provided in this **Memorandum of Agreement and Informed Consent for Research Participation**.

I have been given the opportunity to ask questions regarding the study, the prototype system, and my participation. I understand the purpose of the study, the activities that I may be asked to perform, the expected duration, the possible inconveniences and risks, the potential benefits, and how information collected during the study will be used.

I understand that participation in the research evaluation activities is voluntary and that I may refuse to participate or withdraw from participation at any time without penalty.

I understand that the system is an experimental prototype developed for academic research and that its performance, accuracy, and reliability are not guaranteed.

By signing below, I freely agree to participate in the research activities described in this document.

I understand that I will be provided with a copy of this document for my records.

SIGNATURES

Participant / Client

Name: Lorenz Go Lim

Signature: _____

Date: October 11, 2026

Researchers

Paul Andrew F. Parras
Signature: _____
Date: October 11, 2026

Vaun Michael C. Pace
Signature: _____
Date: October 11, 2026

Lorenz Anthony L. Soriano
Signature: _____
Date: October 11, 2026

Noted by:

Engr. Alvin Joseph S. Macapagal
Thesis Adviser
Engr. Jan Dave Q. Campañera
Assistant Thesis Adviser

Appendix B
Project Cost Estimate

Proposed Budget for Materials

Item Description	Quantity	Cost/Unit (PHP)	Subtotal (PHP
[1] Raspberry Pi 4/5 (Central Controller)	1	₱7,449.00	₱7,449.00
[2] ESP32 Microcontroller (Edge Feeder Node)	1	₱450.00	₱450.00
[3] 7-inch DSI Touchscreen Display	1	₱2,800.00	₱2,800.00
[4] 0.18 m3 Hopper & Trough Fabrication	1	₱5,000.00	₱5,000.00
[5] Sensors & Actuators (Load Cell, Flow, MQ-135, Solenoids)	1 set	₱2,500.00	₱2,500.00
Total			**₱18,249.00**

Proposed Budget for Lab Equipment/Apparatus Use

Item Description	Quantity	Cost/Unit	Subtotal
[1] Reference Weights (Phase 1 Testing)	1 set	₱2,000.00	₱2,000.00
[2] MQ-135 Calibration Environment	1	₱1,500.00	₱1,500.00
Total			**₱3,500.00**

Room Rental Costs

Room	No. of hours	Cost/hour	Subtotal
[1] University Engineering Lab	—	₱3,200.00	₱3,200.00
[2] Piggery Field Site (Site Assessment Phase)	—	Provided by client	—
Total			**₱3,200.00**

Miscellaneous Costs

Manpower/Overtime Costs

Name of Person	Position	Rate/Fee	No. of Hours
[1] Manpower (Overtime Installation)	Lead/Helper	₱1,000.0/person	—
Total			**₱3,000.00**

External Analysis

Description	Samples	Analysis Fee	Subtotal

[1] Ammonia Threshold Calibration (Known Concentrations)	1 Session	₱1,500.00	₱1,500.00
Total			**₱1,500.00**

Documentation Costs

Description	Quantity	Cost/Unit	Subtotal
Thesis Manuscript Printing and Binding	5	₱300.00	₱1,500.00
Total			**₱1,500.00**

Transportation and Accommodation Costs

Description	No. of persons	Rate/person	Subtotal
Field Site Visit (Cagayan, Round Trip)	5	₱5,060.00	₱25,300.00
Total			**₱25,300.00**

Incidental Costs

Description	Quantity	Cost/Unit	Subtotal
RS-485 Cables & DC Power Distribution	1 lot	₱1,000.00	₱1,000.00
Total			**₱1,000.00**

GRAND TOTAL	**₱54,249.00**

Appendix C
Workplan and Expected Outputs

WORKPLAN / GANTT CHART

Integrated Automated Feeding and Watering System

TASK / PROCESS	ASSIGNED TO	EXPECTED OUTPUT	DURATION	Weeks													
				1	2	3	4	5	6	7	8	9	10	11	12	13	14
Phase 1: System Architecture Design			2														
1.1 Define system requirements		Requirements doc	1														
1.2 Design hardware & software architecture		Architecture diagram	1														
Phase 2: Hardware Procurement			1														
2.1 Source microcontrollers & sensors		Components acquired	1														
2.2 Source mechanical components		Mechanical parts acquired	1														
Phase 3: Hardware Assembly & Wiring			2														
3.1 Assemble central hopper		Hopper assembled	1														
3.2 Wire feeder nodes		Nodes wired	1														
Phase 4: Software & Firmware Development			2														
4.1 Develop ESP32 firmware		Firmware completed	1														
4.2 Develop Raspberry Pi backend		Backend completed	1														
Phase 5: Database & UI Setup			2														
5.1 Configure SQLite database		Database ready	1														
5.2 Design local touchscreen dashboard		Dashboard UI ready	1														
Phase 6: Hardware Validation & Sensor Calibration			1														
6.1 Gravimetric Testing (FDA Validation)		FDA >= 95%	1														
6.2 Volumetric Testing		Pulse-to-mL factor	1														
6.3 Environmental Baseline		R0 baseline established	1														
Phase 7: Firmware & Communication Testing			1														
7.1 Signal integrity testing		Voltage levels +/- 1.5V	1														
7.2 MODBUS RTU packet loss testing		Packet loss <= 1%	1														
Phase 8: System-Level Integration			2														
8.1 72-hour continuous operation trial		Uptime & Latency <= 2s	1														
8.2 Simulated consumption drops		ARA >= 90%	1														
8.3 LoRa wireless link testing		RSSI & PDR >= 95%	1														
Phase 9: Site Assessment & Electrical Routing			1														
9.1 Cable routing & power plan		Routing diagram & power plan	1														
9.2 EMI risk assessment		EMI risk assessment report	1														
Phase 10: Physical Installation			2														
10.1 Central hopper & module installation		Hopper & module mounted	1														
10.2 Feeder nodes installation		Nodes integrated	1														
10.3 Hardware checklist		Hardware checklist passed	1														
Phase 11: Commissioning & Initialization			2														
11.1 MODBUS device addresses assignment		Addresses assigned	1														
11.2 Silent monitoring phase		Consumption baselines established	1														
Phase 12: Caretaker Turnover & Training			1														
12.1 Touchscreen interface training		Caretakers trained	1														
12.2 Competency evaluation		Competencies demonstrated	1														

Appendix D
Notice of Adviser Acceptance

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NOTICE OF ACCEPTANCE

We, the undersigned, hereby agree to collaborate, as part of the requirements of the Department of Computer Engineering's **BS CPE thesis courses**, in a project/research study entitled, **Integrated Automated Feeding and Watering System with Centralized Data Monitoring for Commercial Piggeries**. This Notice of Acceptance is effective beginning the **1st Semester of AY 2024-2025**.

The **students** will prepare a project/research proposal, including an introductory chapter, a review of related literature, methodology, and a proposed timetable and budget. Upon approval, the students will conduct the study as outlined in the approved methodology, document the results, discussion, conclusions, and recommendations, and defend the study before an appointed thesis evaluation committee.

The **adviser**, whose own research project(s) may provide the foundation for the students' study, is responsible for guiding the students through the project/research study, holding regular consultation sessions at a mutually agreed frequency (e.g., weekly, bimonthly), and reviewing and critiquing the thesis manuscript before it is submitted to the thesis evaluation committee for the proposal hearing and oral examination.

The **assistant adviser** will also support the students within his/her technical capacity and expertise. In the event of the adviser's absence or any circumstance that precludes the adviser from fulfilling their duties, the assistant adviser shall assume the responsibilities of thesis advising.

To formalize this agreement, we hereby affix our signatures below.

PACE, VAUN MICHAEL C.

PARRAS, PAUL ANDREW F.

SORIANO, LORENZ ANTHONY L.

ENGR. ALVIN JOSEPH S. MACAPAGAL
Thesis Adviser

ENGR. JAN DAVE Q. CAMPAÑERA
Thesis Assistant Adviser

Witnesses:

DR. GODWIN S. MONSERATE
CPE 3107 Coordinator

DR. ANTONIETTE M. CAÑETE
Chair, Department of Computer Engineering

USC - Talamban Campus
Gov. M. Cuenco Ave., Talamban, Cebu City, Philippines 6000
Email: comedept@usc.edu.ph
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Appendix E

Student-Adviser Thesis Counseling Logbook

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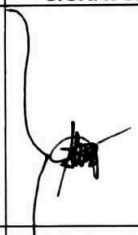
Department of Computer Engineering
School of Engineering

STUDENT-ADVISER THESIS COUNSELING LOGBOOK

✓ Thesis Proposal (CPE 3207L) Thesis Implementation (CPE 3301L | CPE 4101L)

1st Semester AY _____ ✓ 2nd Semester AY 2025-2026 Summer _____

GROUP LETTER:	**V**
PROJECT TITLE:	**Integrated Automated Feeding and Watering System with Centralized Data Monitoring for Commercial Piggeries**
NAME OF ADVISER:	**Engr. Alvin Joseph Macapagal**
NAME OF ASST. ADVISER:	**Engr. Jan Dave Q. Campañera**
NAME OF STUDENTS:	**Vaun Michael C. Pace**
	Paul Andrew F. Parras
	Lorenz Anthony L. Soriano

DATE	VENUE / MEDIUM	TIME	TOPICS DISCUSSED	ADVISER'S / ASST. ADVISER'S SIGNATURE
12/04/2025	DML	10:15 AM - 10:30 AM	-Presented a thesis proposal, and the adviser recommended securing a validating client and identifying a gas sensor suitable for detecting swine emissions.	
12/17/2025	DML	2:00 PM - 3:00 PM	-Discussed with the adviser about conducting a feasibility study for the clients.	

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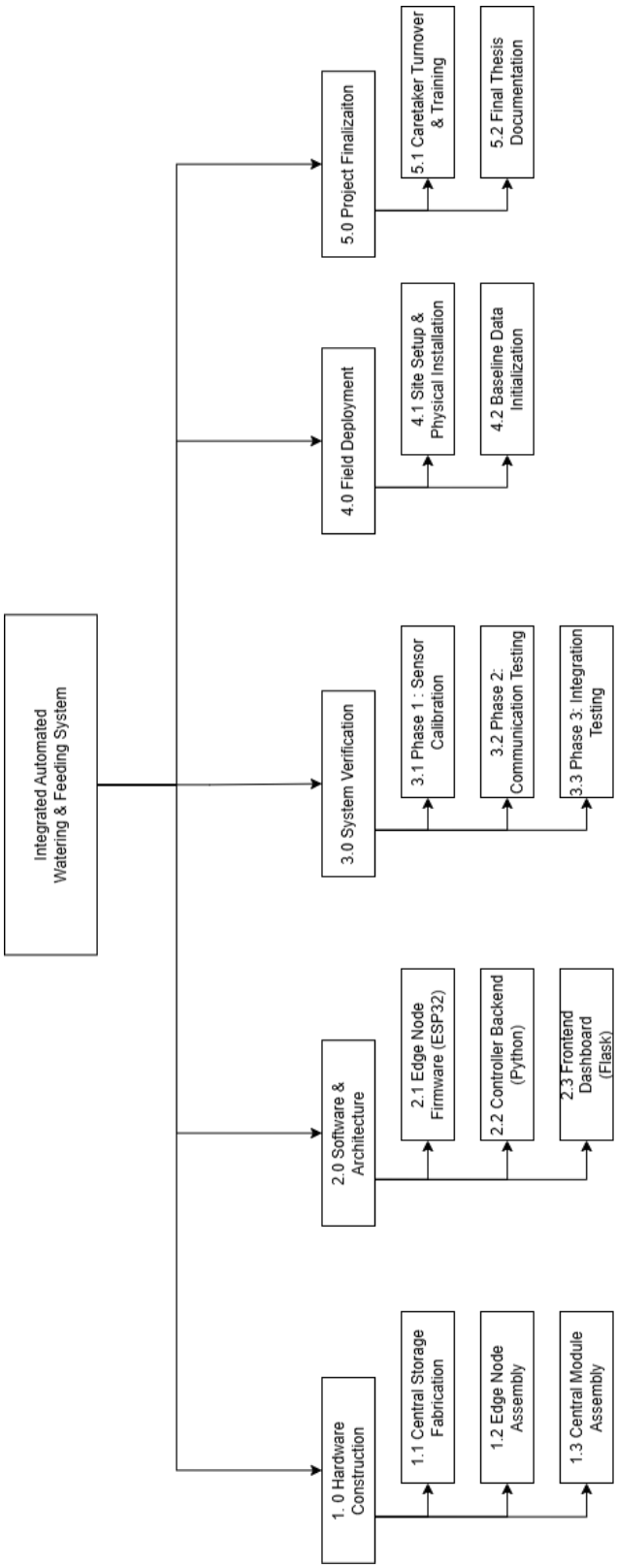
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03/18/2025	DML	1:30 PM - 2:30 PM	-Begin drafting, obtain a thesis copy from Engr. Campañera. -Confirm details with the client -Review related papers to identify common elements.	
03/25/2025	DML	1:30 PM - 2:10 PM	-Clearly qualify the problems, formulate the objectives, structure the manuscript based on the required format. -Focus on completing the Introduction, Significance of the Study, and Goals and Objectives sections.	
04/08/2025	DML	1:30 PM - 2:00 PM	- Develop the methodology -Restructure the goals and objectives -Decide on testing conditions, preferably validating the system with pigs.	
04/15/2025	DML	2:00 PM - 2:15 PM	-changes to the methodology based on the adviser's recommendations	
04/27/2025	DML	1:00 PM - 1:20 PM	-Finalizing the manuscript	

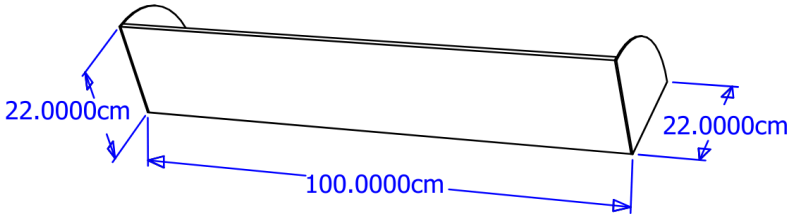
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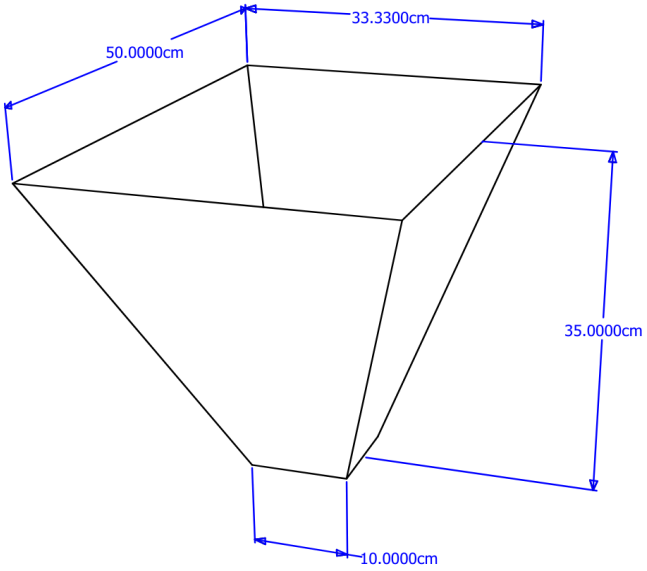
Appendix F
Work Breakdown Structure



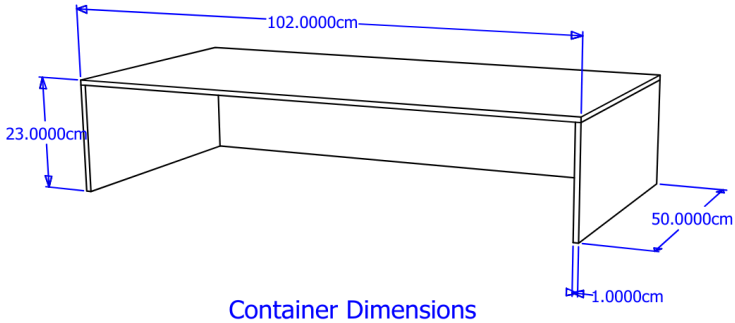
Appendix G
Prototype Design



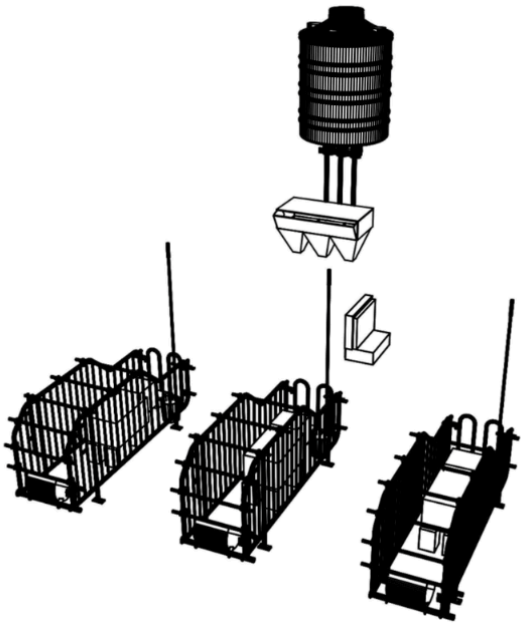
Lid Dimensions



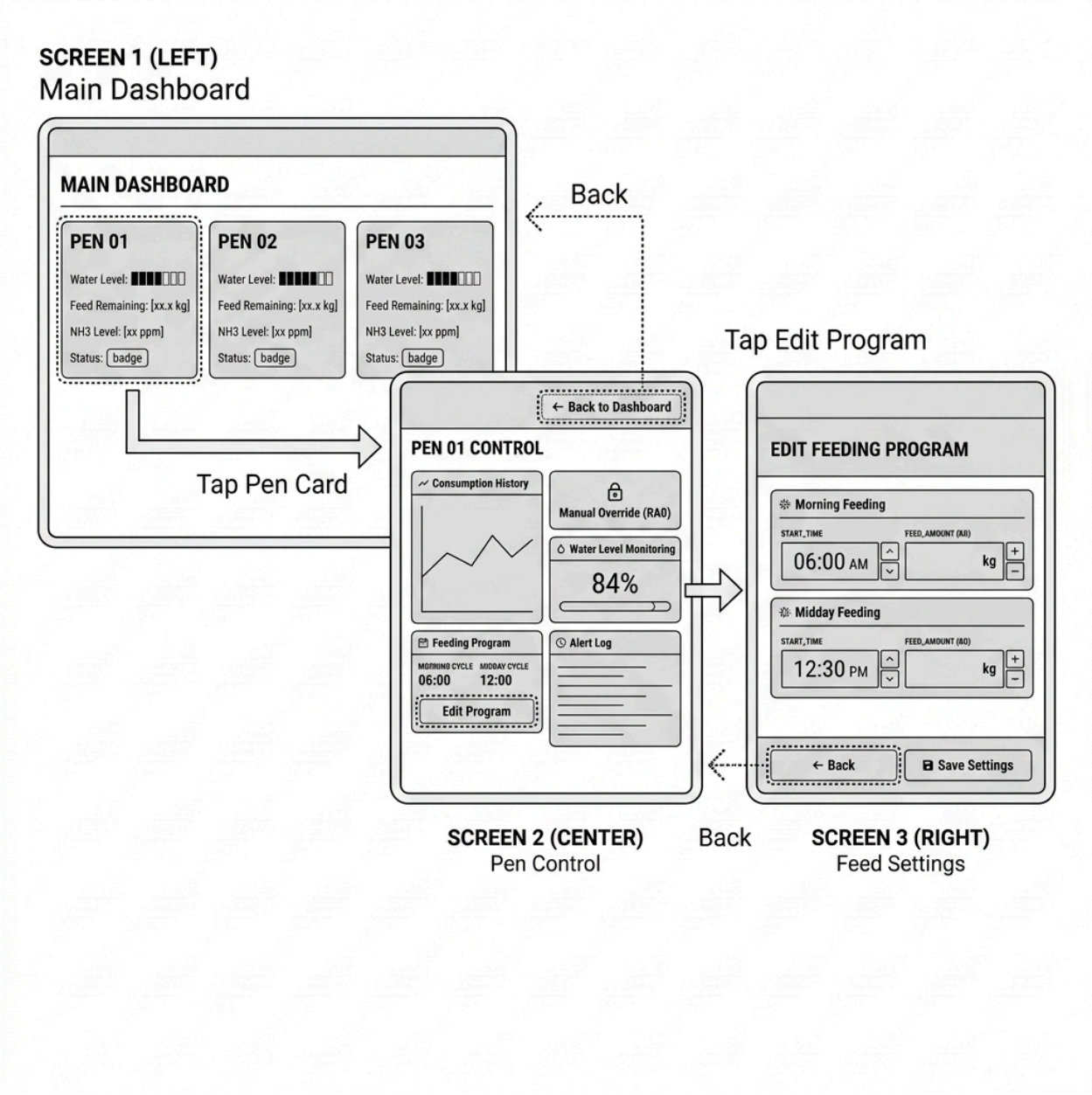
Hopper Dimensions



Container Dimensions



Appendix H
UI Wireframe



Appendix I
Project Deliverables

#	Project Deliverables	Status
A. Hardware Deliverables		
1	Central Control Module (Raspberry Pi, Touchscreen, RS-485, LoRa, UPS)	Ongoing
2	Central Feed Hopper Assembly (0.18 m^3 Frustum Hopper with Guillotine Valve)	Ongoing
3	Auger Conveyor Mechanism (NEMA 17 Stepper Motor-Driven Feed Dispenser)	Ongoing
4	Feeder Module Assemblies ×3 (ESP32, HX711, Load Cell, MQ-135, JSN-SR04T)	Ongoing
5	Feed Trough Units ×3 (with Load Cell Mounts)	Ongoing
6	Water Trough & Solenoid Valve Assembly ×3 (PVC-Piped Supply Tank)	Ongoing
7	RS-485 Wired Bus Network (Twisted-Pair, MAX485 Transceivers)	Ongoing
8	LoRa-GSM Remote Alert Gateway	Ongoing
9	UPS Power Backup Module (18650 Lithium Cell Circuit)	Ongoing
10	Manual Override Push Button (Active-High, Pin RA0)	Ongoing
11	Standard Reference Weights (for Load Cell Calibration)	Ongoing
B. Software Deliverables		
12	ESP32 Edge Node Firmware (FreeRTOS Dual-Core, MODBUS RTU Slave)	Ongoing
13	Raspberry Pi Python Backend (PyModbus Poller, Command & Alert Services)	Ongoing
14	Anomaly Detection / Analytical Engine (Rolling-Window, N=14 Sessions)	Ongoing
15	SQLite Local Database (Timestamped Telemetry, Alerts, Schedules)	Ongoing

#	Project Deliverables	Status
16	Flask Web Application Backend (RESTful API, AJAX Polling)	Ongoing
17	Touchscreen Dashboard Frontend (Main Dashboard, Pen Control, Feed Settings)	Ongoing
18	LoRa-SMS Alert Pipeline (Message Queuing, LoRa Payload Encoding)	Ongoing
19	Python Watchdog Service (Heartbeat Logging, Uptime Computation)	Ongoing
C. Testing and Evaluation Deliverables		
20	Feed Dispensing Accuracy Report – T6.1 (FDA ≥ 95%, 30 Trials/Module)	Ongoing
21	Water Level Sensor Validation Report – T6.2 (±1 cm, 10 Trials)	Ongoing
22	MQ-135 Air Quality Calibration Report – T6.3 (25 ppm NH_3 Threshold)	Ongoing
23	RS-485 Signal Integrity Report – T7.1 (Differential Voltage ≥ ±1.5 V)	Ongoing
24	MODBUS RTU Packet Loss Report – T7.2 (Packet Loss ≤ 1%)	Ongoing
25	System Uptime & Data Latency Report – T8.1 (Usys ≥ 95%, Latency ≤ 2 s)	Ongoing
26	Alert Response & SMS Delivery Report – T8.2 (ARA ≥ 90%, SDR ≥ 95%)	Ongoing
27	LoRa Communication Range Report – T8.3 (RSSI ≥ −120 dBm at Max Distance)	Ongoing
28	Load Cell Calibration & Accuracy Report - T2.1 (R^2 ≥ 0.99, ±2% accuracy)	Ongoing
29	End-to-End Alert Latency Report - T4.3 (Mean latency ≤ 30 seconds per alert)	Ongoing
D. Deployment Deliverables		
30	Site Assessment & Electrical Routing (Cable Routing Diagram, EMI Report)	Ongoing
31	Physical Installation (Hopper, Feeder Nodes, Solenoid Plumbing)	Ongoing

#	Project Deliverables	Status
32	Commissioning & Baseline Initialization (MODBUS Addresses, N=14 Baseline)	Ongoing
33	Caretaker Turnover & Training (Dashboard Use, Alert Response, Maintenance)	Ongoing
E. Documentation Deliverables		
34	Undergraduate Thesis Proposal Document (IEEE-Formatted Full Paper)	Ongoing
35	System Architecture & Module Flowchart Diagrams (Figs. 1–5, 11, 14–17, 21)	Ongoing
36	Hardware Engineering Drawings (Figs. 6–9 with Volume Calculations)	Ongoing
37	Dashboard UI Mockups (Main Dashboard, Pen Control, Feed Settings Screens)	Ongoing
38	Testing & Verification Protocol (Tests T6.1–T8.3 Mapped to Objectives)	Ongoing
39	Deployment Plan (4-Phase Field Deployment Strategy)	Ongoing

Appendix J
Researchers' Curriculum Vitae

Lorenz Anthony L. Soriano

PROFESSIONAL SUMMARY

Computer Engineering undergraduate with practical experience in embedded systems, software development, networking, and hardware-software integration. Worked with microcontrollers, Python applications, object-oriented programming, data visualization, AI-assisted tools, and network technologies. Familiar with server administration, network security, microprocessor architecture, and engineering computation. Interested in applying technical and problem-solving skills in embedded systems, software, networking, and other computer engineering fields.

Undergraduate Student, Bachelor of Science in Computer Engineering

PERSONAL DETAILS

Address	Cagayan de Oro City, Misamis Oriental, Philippines
Phone	(+63) 917 726 0812
Mail	lorenzal.soriano@gmail.com
GitHub	github.com/CoolKidd65
Language	English, Cebuano, Tagalog, Mandarin

TECHNICAL SKILLS

Programming	C, Java, Python, Assembly (x86/8086), MATLAB
Mathematics	Calculus, Differential Equations, Numerical Methods
Fundamentals	Embedded Systems, Computer Networking and Network Technologies, Server Administration, Network Security Systems, Data Structures and Algorithms, Object-Oriented Programming

PROJECTS

Line Follower Robot | *Embedded Systems* | *PIC16F877A*

- Built a line-following robot using a PIC16F877A microcontroller, sensors, and motor control.

Electricity Bill Projector | *Python, Streamlit*

- Built a web application that estimates electricity usage and projects monthly bills using Lagrange polynomial extrapolation.

Custom Multi-Rail DC Power Supply | *Hardware Project*

- Designed and built a multi-rail DC power supply with variable output and overcurrent protection.

Parking Management System | *Object-Oriented Programming* | *Java*

- Developed a Java-based parking management system using object-oriented programming principles.

CIFAR-100 Image Classification Project | *Deep Learning* | *Python, TensorFlow*

- Trained and evaluated a convolutional neural network on CIFAR-100 to improve image classification accuracy.

AI-Assisted Video Clipping Tool | *AI, Video Processing*

- Built a tool that uses AI to identify useful or engaging parts of long-form videos for short-form content.

Copy Trading Chart Normalizer | *Financial Analytics*

- Developed a tool that normalizes trading charts to make trader performance easier to compare.

Personal Finance Tracker | *Data Visualization, Financial Analytics*

- Built a finance tracking application that displays spending habits, trends, and expense summaries using graphs.

AI-Assisted Survey Response Generator | *AI, Automation*

- Built an AI-assisted tool that generates survey responses based on a user-defined profile and page content.

EDUCATION

Oro Christian Grace School 2009 - 2023

Elementary, Junior High School, Senior High School, STEM Strand, Cagayan de Oro City

University of San Carlos 2023 - Present

Bachelor of Science in Computer Engineering, Talamban, Cebu City



Lorenz Anthony L. Soriano

Vaun Michael C. Pace

PROFESSIONAL SUMMARY

Motivated Computer Engineering undergraduate student with comprehensive expertise in microprocessors, digital signal processing, and embedded systems development. Demonstrated proficiency in hardware-software co-design, control systems, and full-stack software application development through hands-on technical projects. Aspiring to leverage technical competencies in software engineering and embedded intelligence to contribute to innovative hardware-software solutions and advance full-system development initiatives.

Undergraduate Student, Bachelor of Science in Computer Engineering

PERSONAL DETAILS

Address	Mandaue City, Cebu
Phone	(+63) 961 241 0607
Mail	pacevaun@gmail.com
GitHub	github.com/kimchaewon34
Language	Filipino, English, Cebuano

TECHNICAL SKILLS

Programming	C, C++, Java, Python, Assembly (8086), MATLAB
Mathematics	Numerical Methods, Discrete Mathematics, Calculus, Advanced Mathematics
Fundamentals	Object-Oriented Programming, Data Structures and Algorithms, Digital Signal Processing, Microprocessors
Tools	Quartus, Proteus, KiCAD, MATLAB, Arduino IDE, Tinkercad, VSCode, CodeBlocks, Git/GitHub

PROJECTS

AC Control System | *Microprocessors* | *8086 Assembly, Proteus*

- 8086-based AC controller with thermostat and timer modes, fan and swing control, and motor drivers.

Line Follower Robot | *PIC16F877A*

- Line-following robot built on a PIC16F877A microcontroller.

Ultrasonic Distance Measurement with DSP | *Digital Signal Processing* | *Arduino Mega, C++*

- 10-point moving average FIR filter on HC-SR04 readings, plotted raw vs. filtered in real time.

Electricity Bill Projector | *Programming Project* | *Python, Streamlit*

- Web app that projects month-end kWh and bill amount using Lagrange extrapolation.

SDN Traffic Monitor with ANN | *Python, Scapy, TensorFlow*

- ANN-based flood traffic detector that triggers OpenFlow block rules, with an authenticated sniffer-to-monitor handshake.

ADDmeCART | *Symfony, Twig, SQLite*

- E-commerce catalog and inventory management web app.

Custom Power Supply

Parking Lot System | *Java*

- Java-based parking lot system with a modular muti-package design.

CONFERENCES

RACCETCon 2026 March 12-13, 2026

Robotics and Automation, Computer Engineering and Technology, and Sustainable Energy Ecosystems Conferences. BAI Hotel, Mandaue City, Cebu

EDUCATION

University of San Carlos 2023 - Present

Bachelor of Science in Computer Engineering, Talamban, Cebu City

Academia de San Jose 2021 - 2023

Senior High School, STEM Strand, Mandaue City



VAUN MICHAEL C. PACE

Paul Andrew F. Parras

PROFESSIONAL SUMMARY

Motivated Computer Engineering undergraduate student with strong foundations in embedded systems, microprocessor architectures, and software integration. Proven track record through hands-on projects involving microcontrollers, web applications, and hardware-software systems. Seeking to leverage expertise in hardware design and modern programming frameworks to contribute to innovative engineering solution teams and foster career growth in embedded software development.

Undergraduate Student, Bachelor of Science in Computer Engineering

PERSONAL DETAILS

Address	Fatima, Ubay, Bohol, Philippines
Phone	(+63) 906 394 8224
Mail	parraspaul13874@gmail.com
GitHub	github.com/1ceduu
Language	Filipino, English, Cebuano

TECHNICAL SKILLS

Programming	C, C++, Java, Python, Assembly (8086), MATLAB
Mathematics	Numerical Methods, Discrete Mathematics, Calculus, Advanced Mathematics
Fundamentals	Object-Oriented Programming, Data Structures and Algorithms, Digital Signal Processing, Microprocessors
Tools	Quartus, Proteus, KiCAD, MATLAB, Arduino IDE, Tinkercad, VSCode, CodeBlocks, Git/GitHub

PROJECTS

Line Follower Robot | *Embedded Systems* | *PIC16F877A*

- Line-following robot built on a PIC16F877A microcontroller.

Electricity Bill Projector | *Programming Project* | *Python, Streamlit*

- Interactive web application built using Python and Streamlit that estimates monthly electricity consumption and projects end-of-month bill amounts using Lagrange polynomial extrapolation.

Custom Power Supply | *Hardware Project* | *Linear Power Supply, Transformer, Rectifier, Voltage Regulator*

- Designed and fabricated a multi-rail DC power supply with variable output voltage and integrated overcurrent protection for bench testing.
- Developed a desktop management system applying core object-oriented programming principles, features, and design patterns with complete unit tests and documentation.

TalambanMS | *Thesis Project* | *React, Next.js, Firebase, Gemini Flash API, Polygon, OpenStreetMap*

- Blockchain-secured LGU service hub featuring AI-assisted document processing via Gemini API, GCash integration, and OpenStreetMap business directory.

CONFERENCES

RACCETCon **2026** March 12-13, 2026

Robotics and Automation, Computer Engineering and Technology, and Sustainable Energy Ecosystems Conferences. BAI Hotel, Mandaue City, Cebu

EDUCATION

University of San Carlos 2023 - Present
Bachelor of Science in Computer Engineering, Talamban, Cebu City

Saint Paul's School Foundation Incorporated 2017 - 2023
Junior Highschool, Senior Highschool, STEM Strand, Ormoc City



PAUL ANDREW F. PARRAS

Appendix K
Advisers' Curriculum Vitae

ALVIN JOSEPH S. MACAPAGAL

Assistant Professor | Computer Engineering

+63 991 608 4686 • alvinjoseph.macapagal@usc.edu.ph • Cebu City, Philippines

EDUCATION

Master of Engineering in Computer Engineering | University of San Carlos – Talamban Campus
Major in Digital Systems Design • October 2015

Bachelor of Science in Computer Engineering | University of San Carlos – Talamban Campus
Major in Digital Systems Design • March 2009

PROFESSIONAL EXPERIENCE

Faculty | University of San Carlos, Department of Computer Engineering
June 2009 – Present

Laboratory-in-Charge | Digital and Microprocessors Laboratory, University of San Carlos
November 2015 – Present

Administrative Head | Printed Circuit Board Prototyping Laboratory, University of San Carlos
June 2011 – May 2012

Operations Head | Printed Circuit Board Prototyping Laboratory, University of San Carlos
June 2011 – May 2012

PROFESSIONAL DEVELOPMENT

- 5th International | 13th National Convention and General Assembly of the Institute of Computer Engineers of the Philippines — November 19–21, 2025
- CIDR Program Planning and Mirror Laboratory Workshop — May 15, 2025
- Sustainable Energy Ecosystems International Conference (SEECON) 2025 — February 6–7, 2025
- 57th AI3 and SOI Asia Directors Meeting — June 20–21, 2024
- GRI Certified Sustainability Professional Training & Carbon Accounting Workshop — March 8, 14–16, 2024
- Minebea Mitsumi Robotics Engineering Immersion Program — August 5–10, 2019
- University of San Carlos 4th University Research Conference — May 2–3, 2019
- 15th DOST-ERDT Conference and 11th AUN/SEED-Net Regional Conference on Energy Engineering — September 27–28, 2018
- 3rd International Symposium on Mobile Internet Security (MobiSec 2018), Local Arrangement Committee Member — August 31, 2018
- Robotics and Automation Conference and Computer Engineering and Technology Conference 2018 — July 13, 2018
- Basic Occupational Safety and Health (BOSH) — April 11–14, 2018
- University of San Carlos 3rd University Research Conference — March 15–16, 2023
- 14 DOST-ERDT Conference, 2017 AEESEAP Workshop & 27th Executive Committee Meeting, ICSEE International Conference on Sustainable Energy Ecosystems — October 18–20, 2017
- International Engineering Research Conference — July 19–21, 2017
- RoboFESTA-PH 2016, Event Manager — December 4, 2016
- Philippine Association of Extension Program Implementors, Inc. 5th Biennial Convention and 1st International Conference — October 26–28, 2016
- RoboFESTA-PH 2013, Event Organizer — March 10, 2013
- RoboFESTA-PH 2010, Event Organizer — March 7, 2010
- Job Preparation Symposium — December 18, 2008
- Understanding Inkjet Technology — November 29, 2008
- Introduction to Electromagnetic Compatibility — November 29, 2008
- Introduction to LabVIEW — November 29, 2008
- International Engineering Research Conference 2017 — July 2017
- OBE Regional Conference — November 2016
- USC University Research Conference — March 2016
- SenseDev'16 — March 2016
- Workshop on Sustainable Energy Ecosystems — February 2016
- Training Workshop on Geographic Information System (GIS) Techniques — December 2015
- ICpEP National Convention 2015: Enhancing Competencies and Meeting Future Challenges — November 2015

- Robotics and Automation Conference – Computer Engineering and Technology Conference (RAC-CETCon) — September 2014
- Integrated CASEE Seminar–Workshop — May 2014
- USC Workshop for Laboratory Personnel — May 2012
- Solar-Energy: Photovoltaic (PV) System Design — December 2011
- Embedded Systems Development and Applications Workshop — January 2011
- Pollution Control Officer Seminar Workshop — July 2010
- International Conference on Electronics, Information, and Communication — June 2010
- Engineering Design Using SolidWorks — June 2010
- Workshop on PCB Design and Layout Using Altium — January 2010
- HRMO–IDU Seminar — September 2009
- Robotics and Automation Conference (RAC) — March 2009

RESEARCH PROJECT INVOLVEMENT

- Implementation of an Enhanced Patient Information System for the Health Services Department of USC — January 2025–May 2026
- Integrated RFID Sticker System for Seamless Vehicle Identification – An Innovative Solution for Contactless Campus Access — January 2025–May 2026
- Sustainable Vehicle Development: Enhancement of E-Plete System — August 2024–May 2025
- Development of a Web-based Smart Television Content Management System — January 2024–May 2025
- Parking Space Monitoring and Navigation Guide System Through Mobile Application and Sensor Technology — January 2024–May 2025
- Investigation of Deep Learning for Predicting and Simulating Plant Growth Structures in Vertical Farming — January 2024–May 2025
- Redevelopment and Evaluation of an IoT Urban Farming System Using ESP32 with Sensors, Cloud Data Integration and Mobile App Monitoring — January 2024–May 2025
- An Automated Inventory Management for an Egg Production Business in the Philippines — January 2024–May 2025
- Non-Intrusive Coconut Maturity Characterization Using Ultrasonic Wave Analysis with Machine Learning-Based Maturity Level Identification — January 2024–May 2025
- Automated Monitoring for SDN-Based Secure Internet-Enabled Video System with Machine Learning Approach — January 2022–May 2023
- Development of Secure Specification-Based Misbehavior Detection Application for Artificial Pancreas System — January 2022–May 2023
- An IoT-Based Ambient Air Quality Monitoring System for Urban Cities — August 2019–May 2021
- Automated Monitoring and Management System for Aquaculture Applications — August 2019–May 2021
- Classroom Attendance Monitoring System Using Radio Frequency Identification with Web Application — August 2019–May 2021
- Interconnected Automatic Weather Stations for a Localized Weather Monitoring System — August 2019–May 2021
- Development of a Traffic Signaling Control System for the City of Cebu: Proposed Alternative Replacement of SCATS — August 2019–May 2021
- Real-Time Drowsiness Detection and Warning System — April 2018–May 2019
- Augmented-Reality Assisted Training for Fire Preparedness with Fire Extinguisher Simulation Support — April 2018–May 2019
- A GPS Guided Topsoil, pH Level and Moisture Regulation System for Agricultural Applications — April 2018–May 2019
- Waste Collection and Notification System with Proximity and Gas Sensor — June 2017–March 2018
- Person Identification Through Gait Analysis — June 2017–March 2018
- A Microcontroller-Aided Electrostatic Filter for Air-Conditioning Units — June 2017–March 2018
- Development of a Wireless Sensor Network Based Monitoring Device for Dengue Outbreak Forecasting — June 2017–March 2018
- Footfall Energy Generator System Using Turbine Mechanism with Wireless Data Acquisition — November 2016–October 2017
- Wireless Liquefied Petroleum Gas Leak Detection System Prototype for Condominiums — November 2016–October 2017
- Development of Eye Blink Detection and Simulation System to Minimize Digital Eye Strain — June 2016–March 2017
- Design and Development of Heat Conversion and Electrical Energy Storage System Using Thermoelectric Generator with Data Acquisition — June 2016–March 2018
- Scene Representation Through Binaural Rendering Using Depth Map — June 2016–March 2017
- Prepaid Water Supply System with Micro Hydro Turbine and Leak Monitoring — June 2015–March 2016
- Implementation of Prepaid Electrical Metering System with Web Application — June 2015–March 2016
- 3D Model Image Generation Using a Stereoscopic Vision Imaging System — June 2015–October 2015
- Stereo Vision Aided Mapper Mobile Robot — June 2012–March 2013
- Development of Learning Module for LAPIS 8-bit Microcontroller for Watch Development — June 2012–March 2013
- Autonomous Mapper Mobile Robot — June 2011–March 2012
- Radio Frequency Identification Card Login System with Face Recognition — June 2010–March 2011
- Microcontroller-Based Chassis-Mounted Camera Ball Collecting Robot Using Image Processing and Sensor — June 2010–March 2011
- Implementing the Lynx-5 Robotic Arm Using Denavit-Hartenberg Motion Convention as a Teaching Aid for Robotics — June 2009–March 2010

- FPGA Based Ball Collecting Robot — June 2009–March 2010
- Implementation of a Two-on-Two Robot Soccer System — June 2008–March 2009

RESEARCH PUBLICATIONS

- J.D. Aniñon, J.D. Campañera, K.K. Flores, A.J. Macapagal, C.S.K. Otadoy, J.P. Rodas, J.F. Tabaco. "Classroom Attendance Monitoring System Using Radio Frequency Identification with Web Application." Research Briefs on Information & Communication Technology Evolution (ReBICTE), December 2021.
- N. Encienzo, N.M. Enriquez, A. Giangan, L. Labrador, A.J. Macapagal, P.J. Santillan. "FPGA-Based Ball Collecting Robot." International Conference on Electronics, Information and Communication, June 2010.
- M.K. Abello, M.V. Chua, A. Legaspi, A.J. Macapagal, C.N. Meneses, C. Ongluico. "Implementing the Lynx-5 Robotic Arm Using Denavit-Hartenberg Motion Conventions as a Teaching Aid for Dynamics." International Conference on Electronics, Information and Communication, June 2010.

CERTIFICATIONS

- TESDA NC II – Computer Hardware Servicing
- Basic Occupational Safety & Health (BOSH)

SKILLS & QUALIFICATIONS

- C / C++ Programming
- Printed Circuit Board Design
- HDL (Verilog and VHDL)
- Embedded Systems Programming
- Computer / Machine Vision Applications
- Python and MATLAB Programming

Certification

I hereby certify that the information provided in this curriculum vitae is true and correct to the best of my knowledge.

ALVIN JOSEPH B. MACAPAGAL

Jan Dave Q. Campañera

Mobile No: (+63)9954522144 **Email Address:** jd_campanera@yahoo.com

SKILLS AND CAPABILITIES

Personal Skills

- Speaks English, Filipino, Cebuano
- Critical Thinker
- Resourcefulness
- Perceptiveness
- Collaborative
- Amiable

Technical Skills

- C Programming
- IC Design
- OpenCV
- PLC programming
- 3D printing
- HDL Designer
- Verilog
- Excel VBA
- C++ Programming
- PCB Design
- Microcontroller
- Microsoft Offices
- Soldering
- HTML & CSS
- JavaScript
- Basic Database
- Xamarin

EXPERIENCE

Aug 2023 – Present	Faculty (Part-Time) Department of Computer Engineering University of San Carlos Nasipit, Talamban, Cebu City, Cebu Philippines 6000
May 2022 – Aug 2024	Research Associate – Project Technical Assistant V Center for Geoinformatics and Environmental Solutions (CenGES) University of San Carlos Nasipit, Talamban, Cebu City, Cebu Philippines 6000
Jun 2016 – May 2021	Faculty (Full-Time) Department of Computer Engineering University of San Carlos Nasipit, Talamban, Cebu City, Cebu Philippines 6000
Apr 2015 – Jun 2015	On-Site Technician (Intern) Watchdog Automation Sales and Services Basak, Lapu-Lapu City, Cebu Philippines 6015
Oct 2014 – Nov 2014	Trainee (Full Time) Philippine Institute for Integrated Circuits University of San Carlos, Talamban, Cebu City, Cebu Philippines 6000

09/07/2026

EDUCATIONAL BACKGROUND

June 2011 – May 2016 — Undergraduate/College
BS Computer Engineering Major in Digital System Design
University of San Carlos
Nasipit, Talamban, Cebu City, Cebu Philippines 6000

PROJECTS

- **SMART TUrF Project 3: POSTE (Interconnected Poste Kits for Environmental Sensing)**
  Research Associate – Information Systems Researcher II, May 2022 – Aug 2024
- **Automated Data Gathering System using Government-issued Identification cards for Contact Tracing**
  Co-Adviser, 2020 – present
- **Centralized Control System for Appliances in Audio-Visual**
  Co-Adviser, 2019 – 2020
- **Classroom Attendance Monitoring System Using Radio Frequency Identification with Web Application**
  Co-Adviser, 2019 – 2020
- **Eczema-Type Recognition System Using Image Processing and Neural Network**
  Co-Adviser, 2018 – 2019
- **Visual Surveillance Applying Event Detection for Proper Waste Disposal Monitoring**
  Co-Adviser, 2018 – 2019
- **Augmented-Reality Assisted Training for Fire Preparedness with Fire Extinguisher Simulation Support**
  Co-Adviser, 2018 – 2019
- **Mosquito Trapping Device Through Combination of Artificial Wing-Beat Sound and Yeast Generator**
  Co-Adviser, 2017 – 2018
- **Control System for Hybrid Hot Air Balloon**
  Co-Adviser, 2017 – 2018
- **Monitoring The Feeder and Drinker Visits of Pigs Using Image Processing**
  Co-Adviser, 2017 – 2018
- **An Interactive Project Trail for Metro Cebu Development and Coordinating Board (MCDCB)**
  Researcher (Documenter & Internal Relations), 2015 – 2016

TRAININGS, SEMINARS & AWARDS

- **7th National Convention, National CpE Challenge: Quiz Bowl 2nd Place (ICpEP)**
  Coach, Nov 2019

09/07/2026

- **G Suite End User Training (Kollab & University of San Carlos)**
  Trainee, May 2017
- **RoboFesta-PH 2016, Run Robot Run event (Department of Computer Engineering)**
  Committee Member, Dec 2016
- **4th Mega Cebu Month, IoT Challenge 2nd runner-up (MCDCB)**
  Team Adviser of Project: Antizeus (Flood Monitoring), Sep 2016
- **24th CebuCon Workshop on Sustainable Matters**
  Participant, Jun 2016
- **International Workshop on Sustainable Energy**
  Participant, Feb 2016
- **Philippine Information Technology (Phil-IT) General Certification Examination**
  Participant, Jan 2016
- **Intellectual Property Seminar**
  Participant, Jan 2016
- **Design Thinking Workshop**
  Participant, Oct 2015
- **Inspire Conference 2015 & Sikat Design Challenge 2015**
  Participant: Team Entry, 2015 – 2016
- **Synthesis of Digital Circuits and RTL to GDSII Backend Design**
  Trainee, Oct 2014



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